



A decomposition clustering ensemble learning approach for forecasting foreign exchange rates

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ABSTRACT

A decomposition clustering ensemble (DCE) learning approach is proposed for forecasting foreign exchange rates by integrating the variational mode decomposition (VMD), the self-organizing map (SOM) network, and the kernel extreme learning machine (KELM). First, the exchange rate time series is decomposed into N subcomponents by the VMD method. Second, each subcomponent series is modeled by the KELM. Third, the SOM neural network is introduced to cluster the subcomponent forecasting results of the in-sample dataset to obtain cluster centers. Finally, each cluster's ensemble weight is estimated by another KELM, and the final forecasting results are obtained by the corresponding clusters' ensemble weights. The empirical results illustrate that our proposed DCE learning approach can significantly improve forecasting performance, and statistically outperform some other benchmark models in directional and level forecasting accuracy.

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1. Introduction

Rapid development of China's economy has resulted in quick growth in international use of the Renminbi (RMB). According to a report from the Society for Worldwide Interbank Financial Telecommunication, the RMB accounted for 1.68% of global payments and was the 6th most active currency in December 2016.¹ In November 2015, the International Monetary Fund (IMF) announced the launch of the new Special Drawing Rights (SDR) valuation basket which includes the Chinese RMB. Becoming effective on October 1, 2016, the RMB officially joined the IMF's SDR basket with a weighting of 10.92%, along with the USD, the EUR, the JPY, and the GBP. As China's economy expands, international influence of the RMB is increasing rapidly. Accurate forecasting of exchange rate trends and scientific analysis of exchange rate changes are of great significance for

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¹ <https://www.swift.com/our-solutions/compliance-and-shared-services/business-intelligence/renminbi/rmb-tracker>.

investors and policymakers in order to make strategies and hedge risk. The nonlinearity and high complexity of exchange rate data makes exchange rate forecasting a challenging task. Development of a more effective forecasting approach to improve forecasting performance is very important in both theoretical and practical aspects.

According to the existing literature, a wide variety of methods have been proposed, which can be divided into three categories: econometric and statistical methods, artificial intelligence (AI) techniques, and ensemble learning approaches.

Econometric and statistical methods are widely proposed to forecast exchange rates, including autoregressive integrated moving average (ARIMA) models (Chortareas, Jiang, & Nankervis, 2011), generalized autoregressive conditional heteroscedasticity (GARCH) models (Cai, Chen, & Fang, 2012; Chortareas et al., 2011; Wan et al., 2016), error correction models (ECM) (Moosa & Vaz, 2016), vector auto-regression (VAR) (Carriero, Kapetanios, & Marcellino, 2009; Jiang et al., 2018), Bayesian theory (Byrne, Korobilis, & Ribeiro, 2016) and so on McCrae et al. (2002). employed the ARIMA model with error correction to forecast daily Japanese Yen, Malaysian Ringgit, Philippines Peso, Thai Baht, Singapore Dollar against the USD Cai et al. (2012). designed a GARCH-type model with a policy dummy variable to consider the conditional mean and volatility of USD/CNY exchange rate. The econometric and statistical methods perform better with stationary data. However, they have difficulties in capturing the nonlinear hidden patterns in exchange rates, and the complex interconnected relationship between the exchange rate and other economic factors.

To deal with nonlinearity and unstationary data, some complex artificial intelligence (AI) techniques are used for exchange rate forecasting, including artificial neural networks (ANNs) (Galeshchuk, 2016; Hu et al., 1999; Sermpinis et al., 2012; Yu, Wang, & Lai, 2010), grey theory (Li et al., 2012), and deep learning techniques (Shen, Chao, & Zhao, 2015) Shen et al. (2015). improved the classical deep belief network (DBN) model by using continuous restricted Boltzmann machines to construct a DBN to forecast the weekly GBP/USD, BRL/USD exchange rate series and weekly INR/USD exchange rate return series. The above non-linear AI methods have better predictive performance than traditional econometric and statistical models, but also have many problems, such as parameter optimization and overfitting.

Therefore, exploring a more effective predictive model with sufficient learning ability is necessary for exchange rate forecasting. In practice, an individual model can't yield the best performance in all cases. Each model has its own strengths and shortcomings. To overcome the single model's shortcomings and improve the forecasting performance, the ensemble learning approach is proposed, taking advantage of each model's strengths by integrating different approaches (Clarida et al., 2003; Sermpinis et al., 2013, 2015; Sermpinis et al., 2013; Sermpinis et al., 2015; Sun et al., 2017; Wright, 2008; Yang et al., 2017; Yu et al., 2018; Zorzi, Kolasa, & Rubaszek, 2017). More importantly Yu, Wang, and Lai (2008), proposed a decomposition-ensemble learning approach for forecasting of crude oil spot price series. In this learning approach, the original time series is decomposed into many sub-models, which are easier to model and predict than the original time series. The subcomponents are then modeled separately. Finally, the ensemble learning tool is used to generate final results by combining the sub-component forecasts. At present, ensemble learning approaches have been widely used in time series forecasting, such as financial time series forecasting (Plakandaras, Papadimitriou, & Gogas, 2015; Sun et al., 2018), crude oil price forecasting (He, Yu, & Lai, 2012; Yu, Dai, & Tang, 2016; Zhang, Zhang, & Zhang, 2015), and nuclear energy consumption forecasting (Tang et al., 2012).

Based on the principle of "decomposition and ensemble," the ensemble learning approach has proven to be an effective and efficient way to improve forecasting performance. However, combinations of different forecasting methods always have some drawbacks. If the linear combinations are used in ensemble learning approaches to integrate different forecasting methods, the forecast errors may vary over time with characteristics of the components. Hence, linear combinations often lead to a second drawback Sun et al. (2018). confirmed that the self-organizing map (SOM) neural network (NN) can significantly improve the forecasting ability of the proposed clustering-based nonlinear ensemble (CNE) learning approach. Therefore, the decomposition clustering ensemble (DCE) learning approach is proposed in this study, which employs clustering strategy to cluster the sub-component forecasting results. By using different ensemble weights in different forecasting times, better performance can be obtained in comparison to the fixed ensemble weights.

A DCE learning approach to forecasting exchange rates is proposed in this paper by using the variational mode decomposition (VMD), the SOM NN, and the kernel extreme learning machine (KELM). First, an efficient decomposition method named VMD is used to divide the original time series into several relatively independent subcomponents. Second, KELM is used to forecast each component. Third, the SOM NN is employed as the method to cluster the sub-series forecasting results in order to obtain cluster centers. Fourth, another KELM is employed to estimate the ensemble weight of each cluster, and then the corresponding clusters' ensemble weights are used to obtain the final ensemble forecasting results. To verify the proposed DCE approach, it is used to forecast daily exchange rates of the USD against three other main currencies: the EUR, the CNY, and the JPY.

The main contributions of this paper are twofold. First, differing from the existing DCE learning approach (VMD), SOM NN, and KELM are integrated into the decomposition ensemble learning paradigm to obtain a better forecasting power. Second, a novel ensemble learning approach is proposed for forecasting exchange rates. The results of empirical study also show that the forecasting ability of the proposed DCE approach for exchange rates forecasting is better than that of the benchmarks.

The rest of this paper is organized as follows. The related methods are briefly introduced in Section 2. Section 3 gives the formulation of the DCE learning approach. The empirical results and the performance of our proposed approach are discussed in Section 4, and Section 5 concludes this paper.

2. Methodology

Before presenting our new approach, the VMD, SOM, and KELM used in the approach are described briefly in this section.

2.1. Variational mode decomposition

The VMD is a novel signal filtering method originally introduced by Dragomiretskiy and Zosso (2014). In comparison to the traditional Empirical Mode Decomposition (EMD) algorithm, it has strong robustness for sampling and denoising. VMD is a completely non-recursive decomposition method in which the modes are extracted simultaneously. The model looks for a set of patterns and their respective center frequencies such that the patterns collectively reproduce the input signal, and each mode is smooth after demodulation to baseband.

Each mode μ_k is required to be mostly compact around a center pulsation ω_k determined along with decomposition. Moreover, VMD requires the sum of the bandwidth of each mode, which is estimated through the H^1 Gaussian smoothness of the demodulated signal, i.e., the squared L^2 -norm of the gradient is the least, under the constraint that the sum of the modes is equal to the original signal. Hence, for the time series f , the decomposition issue is equivalent to solving the constrained variational issue as follows:

$$\min_{\{\mu_k\}, \{\omega_k\}} \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * \mu_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \quad (1)$$

$$s.t. \quad \sum_k \mu_k = f$$

where $\{\mu_k\} := \{\mu_1, \mu_2, \dots, \mu_K\}$ and $\{\omega_k\} := \{\omega_1, \omega_2, \dots, \omega_K\}$ are shorthand notations for the set of K modes and their center frequencies, respectively. δ denotes the Dirac distribution, t denotes time script, $j^2 = -1$ and $*$ is the convolution operator. To render the problem unconstrained, a quadratic penalty term and Lagrangian multipliers λ are introduced. Then, the augmented Lagrangian function is as follows:

$$\begin{aligned} \mathcal{L}(\{\mu_k\}, \{\omega_k\}, \lambda) := & \alpha \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * \mu_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \\ & + \left\| f(t) - \sum_k \mu_k(t) \right\|_2^2 + \langle \lambda(t), f(t) - \sum_k \mu_k(t) \rangle \end{aligned} \quad (2)$$

where α is the balancing parameter of the data-fidelity constraint. Then, Eq. (2) is addressed by the alternate direction method of multipliers. The mode $\mu_k(\omega)$ in frequency domain is updated by Eq. (3), the updated equation of center frequencies ω_k is provided in Eq. (4), and λ is updated by Eq. (5). The mode $\mu_k(t)$ in time domain is calculated by the real part of the inverse Fourier transform of $\mu_k(\omega)$ in Eq. (3).

$$\hat{\mu}_k^{n+1}(\omega) = \frac{\hat{f}(\omega) - \sum_{i \neq k} \hat{\mu}_i(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k)^2} \quad (3)$$

$$\hat{\omega}_k^{n+1} = \frac{\int_0^\infty \omega |\hat{\mu}_k(\omega)|^2 d\omega}{\int_0^\infty |\hat{\mu}_k(\omega)|^2 d\omega} \quad (4)$$

$$\hat{\lambda}^{n+1}(\omega) = \hat{\lambda}^n(\omega) + \tau \left(\hat{f}(\omega) - \sum_k \hat{\mu}_k^{n+1}(\omega) \right) \quad (5)$$

For further details on VMD method, please refer to Dragomiretskiy and Zosso (2014).

2.2. Self-organizing map network

The SOM network proposed by Kohonen (1990) is a popular ANN-based unsupervised competitive learning algorithm. The SOM network produces a map from a continuous high-dimensional input space Φ to a discredited low-dimensional output space χ . The low-dimensional output space χ consists of q neurons, which are arranged according to some fixed topology. The mapping $c(\chi) : \varphi \rightarrow \chi$ is defined by the weight vectors $W = (w_1, w_2, \dots, w_q)$, and it is assigned to an input vector $x(t)$, a neuron index:

$$i^*(t) = \underset{i}{\operatorname{argmin}} \{ \|x(t) - w_i(t)\| \} \quad (6)$$

where $\|\cdot\|$ refers to the Euclidean distance and t is the current iteration.

A competitive learning rule is applied to the training of the weight vectors. When an input vector is put into the network, the weight vector of the neuron whose weight vector is most similar to the input vector and its neighbors is updated as:

$$w_i(t+1) = w_i(t) + \alpha(t)h(i^*, i; t)[x(t) - w_i(t)] \quad (7)$$

and then the weight vectors of the adapted neurons move slightly towards the input vector. The amount of movement is controlled by the learning rate α , which decreases exponentially with time. The number of neurons affected by this adaption is determined by a neighborhood function h . A common choice of the neighborhood function is the Gaussian function:

$$h(i^*, i; t) = \exp\left(-\frac{\|r_i(t) - r_{i^*}(t)\|^2}{2\sigma^2(t)}\right) \quad (8)$$

where $\|r_i(t) - r_{i^*}(t)\|$ is the distance between neurons i and i^* in the distance output space χ , and $\sigma(t)$ is the radius of the neighborhood function at time t , which decreases exponentially to ensure the reduction of the neighborhood size during training.

The low-dimension output space can show the structure hidden in the high-dimensional data, such as clusters and spatial relationships (Vesanto & Alhoniemi, 2000). Therefore, the SOM is employed to split the historical dataset into several subsets where each subset has its own unique characteristics.

2.3. Kernel extreme learning machine

The extreme learning machine (ELM) proposed by Huang, Zhu, and Siew (2006) is a type of the single-hidden layer feedforward NN (SLFN). The ELM model has been widely applied to many fields owing to its fast learning speed and generalizability. The key highlight of the ELM model is that the input weights and biases are randomly generated, and the hidden layer parameters need not be tuned. The output weights are obtained by simple matrix computations, so the computing time is very short.

For N arbitrary samples (x_i, y_i) $x_i \in \mathbb{R}^N$, $y_i \in \mathbb{R}^N$, $j = 1, 2, \dots, N$. If the activation function of hidden layer is $h(x)$ and the output matrix is Y , then the typical SLFNs can be defined as

$$Y = \begin{bmatrix} y_{1j} \\ y_{2j} \\ \vdots \\ y_{mj} \end{bmatrix}_{m \times N} = \begin{bmatrix} \sum_{i=1}^l \beta_{i1} h(w_i x_j + b_i) \\ \sum_{i=1}^l \beta_{i2} h(w_i x_j + b_i) \\ \vdots \\ \sum_{i=1}^l \beta_{im} h(w_i x_j + b_i) \end{bmatrix}_{m \times N}, (j = 1, 2, \dots, N) \quad (9)$$

Where β represents the network output weights between the hidden layer and the output layer, $w_i = [\omega_{i1}, \omega_{i2}, \dots, \omega_{iN}]^T$ is the input weight between the i^{th} hidden layers and the input layers and l is the number of the hidden nodes; the thresholds of the hidden layer are b . The above equations can also be written as

$$H\beta = Y, \quad Y \in \mathbb{R}^{N \times m}, \beta \in \mathbb{R}^{N \times m}, \quad H = H(\omega, b) = h(\omega x + b) \quad (10)$$

where H is the output matrix of the hidden layer. The input weights and biases are randomly produced instead of being tuned, according to Huang et al. (2006). The only unknown parameter is the output weight β which can be solved by the ordinary least squares method. The solution of the equation above is given by

$$\hat{\beta} = H^\dagger Y, \quad H^\dagger = H^T (HH^T)^{-1} \quad (11)$$

where $H^\dagger$ represents the Moore-Penrose generalized inverse of a matrix H . According to Ridge regression theory and the orthogonal projection method, β can be calculated by adding a positive penalty factor $1/C$ as follows:

$$\hat{\beta} = H^T (1/C + HH^T)^{-1} Y \quad (12)$$

The output function of the ELM can be expressed as

$$f(x) = H\hat{\beta} = HH^T(1/C + HH^T)^{-1}Y \quad (13)$$

The method overcomes some shortcomings of the typical gradient-based learning algorithms, such as overfitting, local minima, and long computation time.

A kernel-based ELM was proposed by Huang et al. (2006). The activation function $h(x)$ of the hidden layer is replaced by a kernel function in terms of Mercer's conditions. The output function of the KELM can be expressed as

$$f(x) = h(x)\hat{\beta} = \begin{bmatrix} k(x, x_1) \\ k(x, x_2) \\ \vdots \\ k(x, x_n) \end{bmatrix}^T (1/C + HH^T)^{-1}Y \quad (14)$$

In this formula, the feature mapping $h(x)$ need not be known to users; instead one may use its corresponding kernel $k(x, x_i)$. This means that a kernel function can replace the random mapping of the ELM, and the output weights are more stable. Therefore, the KELM achieves better generalizability than the ELM. The Gaussian kernel function is employed in this study.

3. DCE learning approach

In this study, the DCE learning approach is obtained by using the VMD as the decomposition method, the SOM NN as the clustering method, and KELM as the forecasting and ensemble tool. The framework of the VMD-SOM-KELM learning approach is illustrated in Fig. 1. The proposed VMD-SOM-KELM learning approach consists of four main steps:

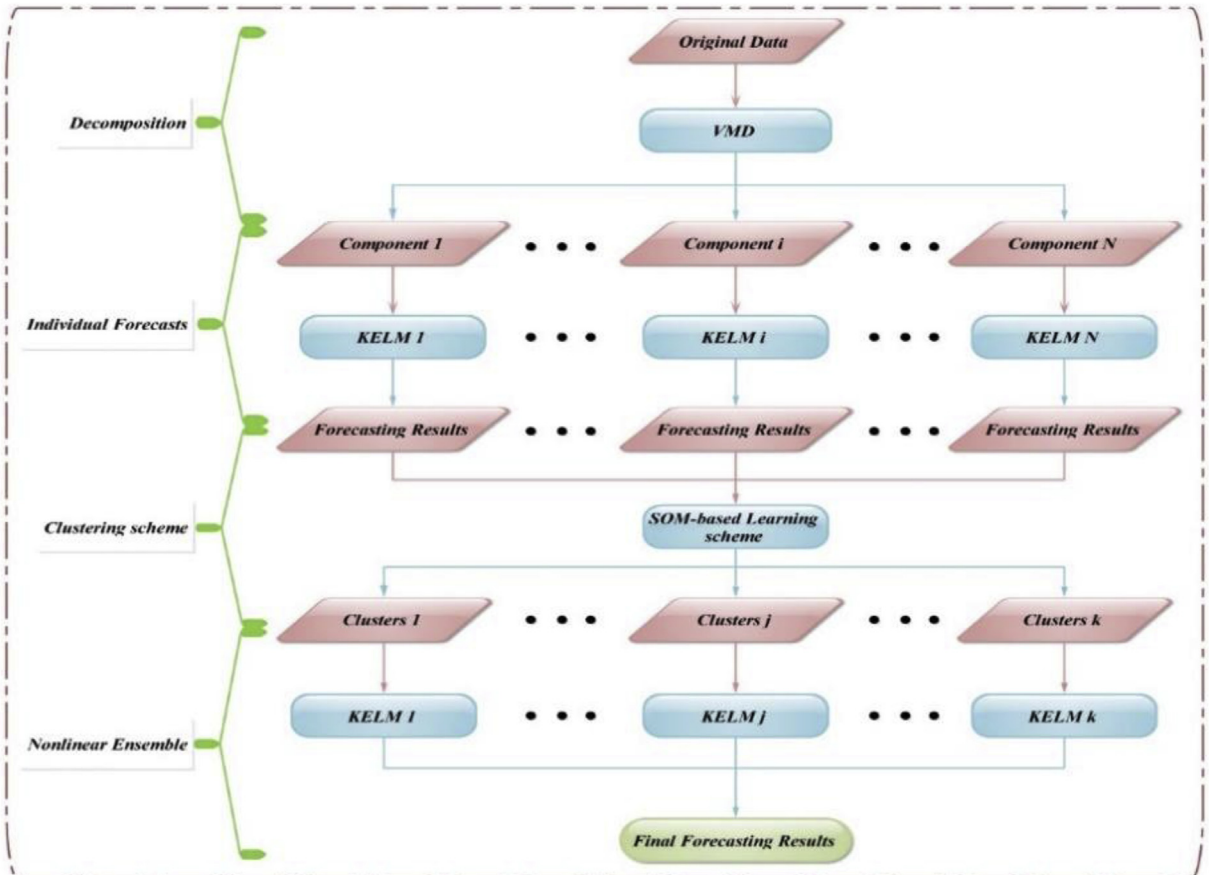


Fig. 1. The framework of the VMD-SOM-KELM learning approach.

- 1) **Decomposition:** The VMD is applied to decompose the in-sample time series $\{y_1, y_2, \dots, y_n\}$ into N subcomponents.
- 2) **Individual forecasting:** Kernel-based ELM is utilized to forecast each subcomponent series separately.
- 3) **Clustering:** The SOM NN is introduced to cluster the subcomponent forecasting results of the in-sample datasets to obtain cluster centers.
- 4) **Ensemble forecasting:** Another kernel-based ELM is employed to estimate each cluster's ensemble weights of in-sample datasets. By comparing the distance between subcomponent forecasting sets and each cluster center, final forecasting results are obtained in terms of the corresponding clusters' ensemble weights.

4. Empirical study

In this section, three major foreign exchange rates are used to verify the forecasting performance of the proposed DCE approach. The description of the data and evaluation criteria are introduced in Section 4.1, and the empirical results are displayed in Section 4.2.

4.1. Data and evaluation criteria

Three daily typical foreign exchange rates include the EUR/USD, USD/CNY, and USD/JPY. The dataset is obtained from the Wind Database (<http://www.wind.com.cn/>), and covers the period of January 3, 2011 to December 29, 2017, with a total of 1820 observations, excluding weekends and holidays. The data set is divided into an in-sample subset and an out-of-sample subset. The in-sample subset is used for model training with data from January 3, 2011 to December 30, 2016. The out-of-sample subset is used for model testing from January 2, 2017 to December 29, 2017. The detailed data are not listed in this paper, but can be accessed from the Wind Database or obtained from the authors.

To verify the forecasting performance of the proposed DCE approach, two main evaluation criteria are adopted to compare the in-sample and out-of-sample forecasting performance: (1) mean absolute percentage error (MAPE), a measure of the deviation between the actual and forecasted values (with smaller values indicating higher forecasting accuracy), is used to evaluate the level forecasting accuracy; (2) directional symmetry (DS), a measure of the performance in predicting the direction of value changes (with higher values indicating better forecasting performance), is used to assess the directional forecasting accuracy. The formulae of MAPE and DS are as follows:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\% \quad (15)$$

$$DS = \frac{1}{n-1} \sum_{i=2}^n d_i \times 100\%, \quad d_i = \begin{cases} 1 & \text{if } (y_i - y_{i-1})(\hat{y}_i - y_{i-1}) \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (16)$$

where $\hat{y}_i$ is the forecasting value, y_i is the actual value and n is the number of observation samples.

Furthermore, the Diebold–Mariano (DM) statistic and Pesaran–Timmermann (PT) statistic are used to evaluate forecasting performances of all models from a statistical perspective (Diebold & Mariano, 2002; Pesaran & Timmermann, 1992). In the DM test, the mean square prediction error (MSPE) is used as the loss function; the null hypothesis of the DM test is that the MSPE of the tested model (te) is not less than that of the benchmark model (be). For further details, please refer to Diebold and Mariano (2002) and Pesaran and Timmermann (1992).

4.2. Empirical results

In this study, four single models (RW, ANN, ELM, and KELM), two decomposition ensemble (DE) learning approaches (EEMD-KELM and VMD-KELM), and one DCE learning approach (EEMD-SOM-KELM) are used as the benchmark models for model comparison.

In order to select model inputs, we use phase space reconstruction to select an optimal input vector of the single NNs. For more detailed information about the phase space reconstruction method, please refer to Takens (1981) and Kim, Eykholt, and Salas (1999). The detailed inputs of the single NNs are presented in Appendix A. The inputs of all KELM-based ensemble learning are similar to single NNs. A group of NNs with different numbers of hidden nodes are trained by means of trial-and-error testing, with each one evaluated by the sum of the square error of the in-sample data set.

Table 1 displays comparisons of forecasting performance of the proposed DCE approach, and the benchmarks for three major exchange rates in terms of level forecasting and directional forecasting via MAPE and DS. The empirical results also demonstrate that:

First, overall, the proposed DCE approach significantly outperforms all benchmark models in terms of level accuracy and directional accuracy for exchange rate forecasting, which verifies that the proposed DCE approach is an effective tool for forecasting foreign exchange rates with the smallest MAPE and the biggest DS.

Table 1

Forecasting performance of daily exchange rates.

Exchange rates	Model type	Competing models	MAPE (%)	DS (%)	Elapsed time (s)	
EUR/USD	Single model	RW	2.182	48.077	0.163	
		ANN	1.325	59.615	7.715	
		ELM	1.089	64.231	2.043	
		KELM	0.794	67.308	3.387	
	DE learning approach	EEMD-KELM	0.547	70.769	14.416	
		VMD-KELM	0.461	72.692	13.297	
	DCE learning approach	EEMD-SOM-KELM	0.345	75.769	17.298	
		VMD-SOM-KELM	0.292	79.231	17.513	
	USD/JPY	Single model	RW	2.969	47.308	0.259
			ANN	1.485	57.692	8.523
ELM			1.154	56.538	1.269	
KELM			0.918	60.769	2.834	
DE learning approach		EEMD-KELM	0.716	64.615	15.214	
		VMD-KELM	0.598	67.692	15.053	
DCE learning approach		EEMD-SOM-KELM	0.602	69.615	18.196	
		VMD-SOM-KELM	0.579	73.846	19.207	
USD/CNY		Single model	RW	2.101	48.361	0.224
			ANN	0.912	53.279	8.135
	ELM		1.059	57.787	2.568	
	KELM		0.812	60.656	3.115	
	DE learning approach	EEMD-KELM	0.283	65.574	13.016	
		VMD-KELM	0.295	64.754	13.293	
	DCE learning approach	EEMD-SOM-KELM	0.214	68.033	16.408	
		VMD-SOM-KELM	0.179	70.082	16.743	

Second, the ensemble learning approach, including the DE learning approach and the DCE learning approach, can obtain better forecasting performance than the single models, including RW, ANN, ELM, and KELM, for all of the three typical foreign exchange rates in terms of MAPE and DS.

Third, for the single models, the KELM outperforms the single models' benchmarks. Since the exchange rate data have the characteristics of high volatility, nonlinearity, complexity, and noise, the KELM (with its strong nonlinearity learning capacity) is more suitable for exchange rate forecasting.

Fourth, the proposed DCE learning approach outperforms the DE learning approach for the three exchange rates in terms of both level accuracy and directional accuracy, which proves that the decomposition process can dramatically improve the forecasting performance of single models.

Fifth, for the DE learning approach, in terms of level accuracy and directional accuracy, forecasting performance of the VMD-KELM is better than the EEMD-KELM; in case of the DCE learning approach, forecasting performance of the VMD-SOM-KELM is better than the EEMD-SOM-KELM, which validates that the VMD algorithm is a more effective decomposition algorithm, with strong robustness to sampling and de-noising.

Furthermore, the DM and PT tests are used to statistically evaluate the level of forecasting performance and the directional forecasting performance of the proposed DCE learning approach, as well as the four single models, two DE learning approaches and another DCE learning approach. The results of the DM and PT tests are displayed in Table 2 and Table 3 respectively.

The DM statistics and p-values of the proposed DCE learning approach (VMD-SOM-KELM) indicate that: (1) compared with the single models, all DM statistics are less than -3.114 , corresponding to p-values of less than 0.000 , which reveals that the proposed DCE learning approach significantly outperforms the single models under 100% confidence level; (2) compared with the DE learning approaches, all DM statistics are less than -2.023 , corresponding to p-values of less than 0.022 , which proves that the proposed DCE learning approach is significantly superior to the DE learning approaches at the three sites under 97.8% confidence level; (3) compared with the other DCE learning approach (EEMD-SOM-KELM), the proposed VMD-SOM-KELM is better than EEMD-SOM-KELM under 83.8% confidence level; and (4) the DCE learning approaches and the DE learning approaches are better than the single models at the three sites under 99.6% confidence level.

Analogously, the PT test is employed to statistically evaluate directional forecasting accuracy. Table 3 shows the results of the PT statistics and the corresponding p-values at the three typical exchange rates. It can be concluded that: (1) the proposed VMD-SOM-KELM DCE learning approach outperforms the benchmarks for the three typical exchange rates with p-value less than 0.001 ; (2) the DCE learning approach is better than DE learning approaches; (3) the VMD-based models outperform EEMD-based models, as VMD is an effective decomposition algorithm.

Compared with the benchmark models for exchange rate forecasting, the proposed VMD-SOM-KELM can obtain better forecasting performance mainly for the following reasons: first, VMD is utilized in the proposed approach to decompose the complex and irregular exchange rate data into N subcomponents. VMD can take advantage of the principle of "decomposition and ensemble" as well as decreasing computational cost when decomposing the original data. Second, the KELM is more suitable to deal with the nonlinearity and unstationary data and is used to forecast subcomponents with different frequencies

Table 2

DM results of daily exchange rates.

Exchange Rates	Models	VMD-SOM-KELM	EEMD-SOM-KELM	VMD-KELM	EEMD-KELM	KELM	ELM	ANN
EUR/USD	EEMD-SOM-KELM	-1.0026 (0.1580)						
	VMD-KELM	-2.0234 (0.0215)	-1.9426 (0.0260)					
	EEMD-KELM	-2.4109 (0.0080)	-2.1369 (0.0163)	-1.8916 (0.0293)				
	KELM	-3.1147 (0.0009)	-3.1023 (0.0010)	-3.0026 (0.0013)	-3.1024 (0.0010)			
	ELM	-3.5726 (0.0002)	-3.4461 (0.0003)	-3.2365 (0.0006)	-3.1789 (0.0007)	-1.0268 (0.1523)		
	ANN	-3.8902 (0.0001)	-3.7054 (0.0001)	-3.7113 (0.0001)	-3.6865 (0.0001)	-1.9906 (0.0233)	-1.5569 (0.0597)	
	RW	-4.5789 (0.0000)	-4.2968 (0.0000)	-4.3028 (0.0000)	-4.2691 (0.0000)	-3.5863 (0.0002)	-3.1426 (0.0008)	-3.5468 (0.0002)
USD/JPY	EEMD-SOM-KELM	-0.9897 (0.1612)						
	VMD-KELM	-2.2309 (0.0128)	-2.0036 (0.0226)					
	EEMD-KELM	-2.6674 (0.0038)	-2.5843 (0.0049)	-1.4416 (0.0747)				
	KELM	-3.5418 (0.0002)	-3.4169 (0.0003)	-2.8984 (0.0019)	-2.8056 (0.0025)			
	ELM	-3.9163 (0.0000)	-3.7463 (0.0001)	-3.1163 (0.0009)	-3.1029 (0.0010)	-1.5127 (0.0652)		
	ANN	-4.1168 (0.0000)	-3.9726 (0.0000)	-3.6258 (0.0001)	-3.5581 (0.0002)	-2.2084 (0.0136)	-1.9685 (0.0245)	
	RW	-4.8812 (0.0000)	-4.5527 (0.0000)	-4.0689 (0.0000)	-3.9682 (0.0000)	-3.6869 (0.0001)	-3.5237 (0.0002)	-3.0125 (0.0013)
USD/CNY	EEMD-SOM-KELM	-1.1053 (0.1345)						
	VMD-KELM	-2.1358 (0.0163)	-2.1058 (0.0176)					
	EEMD-KELM	-2.5369 (0.0056)	-2.6367 (0.0042)	-1.6143 (0.0532)				
	KELM	-3.4815 (0.0002)	-3.3054 (0.0005)	-2.7951 (0.0026)	-2.7014 (0.0035)			
	ELM	-3.7436 (0.0001)	-3.5917 (0.0002)	-3.1058 (0.0009)	-3.0163 (0.0013)	-1.4869 (0.0685)		
	ANN	-4.0918 (0.0000)	-3.9722 (0.0000)	-3.5548 (0.0002)	-3.4476 (0.0003)	-2.3073 (0.0105)	-1.8504 (0.0321)	
	RW	-4.7541 (0.0000)	-4.6021 (0.0000)	-3.9682 (0.0000)	-3.8852 (0.0001)	-3.4527 (0.0003)	-3.3247 (0.0004)	-3.1057 (0.0009)

Table 3

PT results of daily exchange rates.

Model type	Competing models	EUR/USD	USD/JPY	USD/CNY
Single model	RW	0.6729 (0.5010)	0.4176 (0.6762)	0.4259 (0.6702)
	ANN	1.9106 (0.0561)	1.7418 (0.0815)	1.8206 (0.0687)
	ELM	1.9823 (0.0474)	1.8324 (0.0669)	1.9337 (0.0532)
	KELM	2.0314 (0.0422)	1.8906 (0.0587)	1.9769 (0.0481)
DE learning approach	EEMD-KELM	3.1149 (0.0018)	2.9795 (0.0029)	2.7963 (0.0052)
	VMD-KELM	3.3658 (0.0008)	2.9806 (0.0029)	3.0022 (0.0027)
DCE learning approach	EEMD-SOM-KELM	4.2910 (0.0000)	3.5743 (0.0004)	3.7463 (0.0002)
	VMD-SOM-KELM	4.3762 (0.0000)	3.6614 (0.0003)	4.0108 (0.0001)

separately, which can obtain more information than the original data. Thirdly, the SOM NN can significantly improve the forecasting ability of the proposed CNE learning approach (Sun et al., 2018).

5. Conclusions

Because of the intrinsic nonlinearity and complexity of exchange rate data, we propose in this paper a new DCE learning approach for forecasting foreign exchange rates. It integrates the VMD, SOM network, and KELM. Four models, two DE learning approaches and a DCE learning approach are used as benchmark models. The purpose of these models is to verify the

forecasting performance of the proposed VMD-SOM-KELM DCE learning approach with three major exchange rates. The empirical results illustrate that the proposed VMD-SOM-KELM decomposition clustering learning approach significantly outperforms some other popular forecasting methods. This approach outperformed the single models, DE learning approaches without clustering, and another DCE learning approach with EEMD, in terms of level forecasting accuracy and directional forecasting accuracy. These results indicate that the proposed DCE learning approach is a very promising approach for solving complex time series forecasting problems, especially for exchange rate forecasting.

Meanwhile, the proposed DCE learning approach can also be used in some other complex time series forecasting problems, such as stock price forecasting, crude oil price forecasting, container throughput forecasting, and so on.

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Conflicts of interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

Appendix A

Table A1

The results of phase space reconstruction.

Exchange rates	Index	Delay time											
		1	2	3	4	5	6	7	8	9	10	11	12
EURUSD	S_{cor}	0.0349	0.0333	0.0298	0.0370	0.0338	0.0274	0.0211	0.0346	0.0272	0.0366	0.0330	0.0213
	$\tilde{S}(t)$	0.0146	0.0144	0.0128	0.0158	0.0145	0.0122	0.0090	0.0144	0.0113	0.0160	0.0145	0.0089
	$\Delta\tilde{S}(t)$	0.0203	0.0190	0.0171	0.0212	0.0193	0.0153	0.0121	0.0202	0.0160	0.0207	0.0185	0.0124
USDJPY	S_{cor}	0.0390	0.0304	0.0364	0.0296	0.0281	0.0298	0.0363	0.0206	0.0308	0.0242	0.0200	0.0283
	$\tilde{S}(t)$	0.0176	0.0138	0.0167	0.0132	0.0125	0.0135	0.0161	0.0089	0.0142	0.0102	0.0088	0.0126
	$\Delta\tilde{S}(t)$	0.0213	0.0166	0.0197	0.0164	0.0156	0.0163	0.0201	0.0117	0.0166	0.0141	0.0112	0.0157
USDCNY	S_{cor}	0.0915	0.0686	0.0534	0.0519	0.0577	0.0467	0.0439	0.0503	0.0456	0.0351	0.0360	0.0422
	$\tilde{S}(t)$	0.0493	0.0367	0.0278	0.0269	0.0289	0.0241	0.0211	0.0253	0.0222	0.0182	0.0177	0.0203
	$\Delta\tilde{S}(t)$	0.0422	0.0319	0.0256	0.0249	0.0288	0.0226	0.0227	0.0250	0.0234	0.0169	0.0183	0.0219

Note: The bold markers denote the first minimum of $\Delta\tilde{S}(t)$ and the global minimum of $S_{cor}(t)$.

Table A2

The results of the optimal values τ_{ω} , τ and m .

Exchange rates	Max lag	τ_{ω}	τ	m
EURUSD	12	7	3	3
USDJPY	12	11	2	7
USDCNY	12	10	4	4

Table A3

Selected input form by phase space reconstruction.

Exchange rates	Optimal input form	Input size
EURUSD	$x_t^{(1)}, x_{t+3}^{(1)}, x_{t+6}^{(1)}$	3
USDJPY	$x_t^{(2)}, x_{t+2}^{(2)}, x_{t+4}^{(2)}, x_{t+6}^{(2)}, x_{t+8}^{(2)}, x_{t+10}^{(2)}, x_{t+12}^{(2)}$	7
USDCNY	$x_t^{(3)}, x_{t+4}^{(3)}, x_{t+8}^{(3)}, x_{t+12}^{(3)}$	4

References

- Byrne, J. P., Korobilis, D., & Ribeiro, P. J. (2016). Exchange rate predictability in a changing world. *Journal of International Money and Finance*, 62, 1–24.
- Cai, Z., Chen, L., & Fang, Y. (2012). A new forecasting model for USD/CNY exchange rate. *Studies in Nonlinear Dynamics & Econometrics*, 16(3), 1–20.
- Carriero, A., Kapetanios, G., & Marcellino, M. (2009). Forecasting exchange rates with a large Bayesian VAR. *International Journal of Forecasting*, 25(2), 400–417.

- Chortareas, G., Jiang, Y., & Nankervis, J. C. (2011). Forecasting exchange rate volatility using high-frequency data: Is the euro different? *International Journal of Forecasting*, 27(4), 1089–1107.
- Clarida, R. H., Sarno, L., Taylor, M. P., et al. (2003). The out-of-sample success of term structure models as exchange rate predictors: A step beyond. *Journal of International Economics*, 60(1), 61–83.
- Diebold, F. X., & Mariano, R. S. (2002). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 20(1), 134–144.
- Dragomiretskiy, K., & Zosso, D. (2014). Variational mode decomposition. *IEEE Transactions on Signal Processing*, 62(3), 531–544.
- Galeshchuk, S. (2016). Neural networks performance in exchange rate prediction. *Neurocomputing*, 172, 446–452.
- He, K., Yu, L., & Lai, K. K. (2012). Crude oil price analysis and forecasting using wavelet decomposed ensemble model. *Energy*, 46(1), 564–574.
- Huang, G. B., Zhu, Q. Y., & Siew, C. K. (2006). Extreme learning machine: Theory and applications. *Neurocomputing*, 70(1–3), 489–501.
- Hu, M. Y., Zhang, G., Jiang, C. X., et al. (1999). A cross-validation analysis of neural network out-of-sample performance in exchange rate forecasting. *Decision Sciences*, 30(1), 197–216.
- Jiang, Z. H., Deng, P. J., Luo, K. Y., et al. (2018). The effect of market quality on the causality between returns and volatilities: Evidence from CSI 300 index futures. *Journal of Management Science and Engineering*, 3(1), 16–38.
- Kim, H. S., Eykholt, R., & Salas, J. D. (1999). Nonlinear dynamics, delay times, and embedding windows. *Physica D*, 127, 48–60.
- Kohonen, T. (1990). The self-organizing map. *Proceedings of the IEEE*, 78(9), 1464–1480.
- Li, D. C., Chang, C. J., Chen, C. C., et al. (2012). Forecasting short-term electricity consumption using the adaptive grey-based approach—an Asian case. *Omega*, 40(6), 767–773.
- McCrae, M., Lin, Y. X., Pavlik, D., et al. (2002). Can cointegration-based forecasting outperform univariate models? An application to Asian exchange rates. *Journal of Forecasting*, 21(5), 355–380.
- Moosa, I. A., & Vaz, J. J. (2016). Cointegration, error correction and exchange rate forecasting. *Journal of International Financial Markets, Institutions and Money*, 44, 21–34.
- Pesaran, M. H., & Timmermann, A. (1992). A simple nonparametric test of predictive performance. *Journal of Business & Economic Statistics*, 10(4), 461–465.
- Plakandaras, V., Papadimitriou, T., & Gogas, P. (2015). Forecasting daily and monthly exchange rates with machine learning techniques. *Journal of Forecasting*, 34(7), 560–573.
- Sermpinis, G., Dunis, C., Laws, J., et al. (2012). Forecasting and trading the EUR/USD exchange rate with stochastic Neural Network combination and time-varying leverage. *Decision Support Systems*, 54(1), 316–329.
- Sermpinis, G., Stasinakis, C., Theofilatos, K., et al. (2015). Modeling, forecasting and trading the EUR exchange rates with hybrid rolling genetic algorithms—support vector regression forecast combinations. *European Journal of Operational Research*, 247(3), 831–846.
- Sermpinis, G., Theofilatos, K., Karathanasopoulos, A., et al. (2013). Forecasting foreign exchange rates with adaptive neural networks using radial-basis functions and particle swarm optimization. *European Journal of Operational Research*, 225(3), 528–540.
- Shen, F., Chao, J., & Zhao, J. (2015). Forecasting exchange rate using deep belief networks and conjugate gradient method. *Neurocomputing*, 167, 243–253.
- Sun, S., Qiao, H., Wei, Y., et al. (2017). A new dynamic integrated approach for wind speed forecasting. *Applied Energy*, 197, 151–162.
- Sun, S., Wang, S., Wei, Y., et al. (2018). A clustering-based nonlinear ensemble approach for exchange rates forecasting. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*. <https://doi.org/10.1109/TSMC.2018.2799869>.
- Takens, F. (1981). Determining strange attractors in turbulence. *Lecture Notes in Mathematics*, 898, 361–381.
- Tang, L., Yu, L., Wang, S., et al. (2012). A novel hybrid ensemble learning paradigm for nuclear energy consumption forecasting. *Applied Energy*, 93, 432–443.
- Vesanto, J., & Alhoniemi, E. (2000). Clustering of the self-organizing map. *IEEE Transactions on Neural Networks*, 11(3), 586–600.
- Wan, D., Liu, W. Y., Wang, J. B., et al. (2016). Asymmetries of positive feedback trading in individual stocks: Evidences from China. *Journal of Management Science and Engineering*, 1(1), 3–27.
- Wright, J. H. (2008). Bayesian model averaging and exchange rate forecasts. *Journal of Econometrics*, 146(2), 329–341.
- Yang, G. L., Song, Y. Y., Xu, D. L., et al. (2017). Overall efficiency and its decomposition in a two-stage network DEA model. *Journal of Management Science and Engineering*, 2(3), 161–192.
- Yu, L., Dai, W., & Tang, L. (2016). A novel decomposition ensemble model with extended extreme learning machine for crude oil price forecasting. *Engineering Applications of Artificial Intelligence*, 47, 110–121.
- Yu, L., Wang, S., & Lai, K. K. (2008). Forecasting crude oil price with an EMD-based neural network ensemble learning paradigm. *Energy Economics*, 30(5), 2623–2635.
- Yu, L., Wang, S., & Lai, K. K. (2010). *Foreign-exchange-rate forecasting with artificial neural networks*. Springer Science & Business Media.
- Yu, L., Zhou, R., Tang, L., et al. (2018). A DBN-based resampling SVM ensemble learning paradigm for credit classification with imbalanced data. *Applied Soft Computing*, 69, 192–202.
- Zhang, J. L., Zhang, Y. J., & Zhang, L. (2015). A novel hybrid method for crude oil price forecasting. *Energy Economics*, 49, 649–659.
- Zorzi, M. C., Kolasa, M., & Rubaszek, M. (2017). Exchange rate forecasting with DSGE models. *Journal of International Economics*, 107, 127–146.